USACE ERDC CERL COMMERCIAL SOLUTIONS OPENING (CSO 2026) - PRICE & ROM SCHEDULE

Primary Offeror: Humphrey Virtual Farms LLC (CAGE: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5)

Address: 701 E Drew St., Madill, OK 73446 | POC: Jeffery Humphrey, CEO | (572) 358-9978 | humphreyvirtualfarm@gmail.com

Solution Title: Project Ebony v8.0.0: Air-Gapped Tri-Brain OT/SCADA Safety Interlock, Automated Bridge/Pavement NDE & WORM Merkle Provenance Testbed

Contracting Vehicles Supported: Commercial Solutions Opening (CSO) Fixed-Price Agreement, 10 U.S.C. 4022 Prototype OTA (0% Cost Share for NTDC/SDB), or FAR Part 12/13 Commercial Award.

1. Option A: 12-Month Fixed-Price Phase-1 Prototype Effort (TRL 4 to TRL 6) - $1,200,000.00 Total ROM

Humphrey Virtual Farms LLC proposes a 12-Month, $1,200,000.00 Firm-Fixed-Price Milestone Schedule to fabricate, calibrate, and field-validate the TRL 6 CERL Operational Energy & Sustainment Management Prototype (0% Cost Share required per NTDC status):

- Milestone 1 (Months 1-2 | Days 15-30 Mobilization): $150,000.00 ? CERL Operational Energy & Sustainment Management Interface Control Document (ICD), 52-Table SQLite WORM Enclave Baseline & Test Plan.
- Milestone 2 (Months 3-6): $350,000.00 ? Brain 1 Bare-Metal Cortex-M7 + SiC MOSFET Safety Interlock PCB Fabrication & <50us Loop / 36.75ns Cutoff Oscilloscope Verification.
- Milestone 3 (Months 7-9): $350,000.00 ? Brain 2 Isolated Read-Only ASTM D6433 Pavement, Bridge & Facility Sensor Analytics Calibration with ERDC CERL Datasets.
- Milestone 4 (Months 10-12): $350,000.00 ? TRL 6 Integrated CERL Hardware-in-the-Loop (HIL) Demonstration, 110-Control NIST SP 800-171 / 800-82 C3PAO Package & Final Report.
- Total 12-Month Phase-1 Prototype Price: $1,200,000.00 USD (Inclusive of Engineering Labor, PCB/SiC Hardware Fabrication, HIL Test Equipment, and Technical Data Deliverables).

2. Option B: Extended 5-Node Installation Microgrid & SMS Field Pilot OTA - $1,485,000.00 (12 Months)

For installations requiring a multi-bay 5-Node Hardware-in-the-Loop microgrid and Sustainment Management System deployment (matching CDAO Tradewinds Docket 10-26-4046), total 12-month fixed price is $1,485,000.00 ($150,000.00 Mobilization; $445,000.00 Month 4; $445,000.00 Month 8; $445,000.00 Month 12).

3. Option C: Unbundled Dual-Use Commercial Software Licensing Schedule (26-Module C2 Console)

- CLIN 0001 (Base Single-Node TRL 4 Software Testbed & WORM Enclave Pilot): $2,450.00 / Month ($29,400.00 / Year).
- CLIN 0002 through CLIN 0026 (Individual Specialized OT, NDE, Microgrid & Cryptographic Add-On Modules): $20,000.00 to $185,000.00 / Year per module.
- CLIN 1004-A (Full 26-Module 1-Node Tactical Operations Center Bundle): $895,000.00 / Year.
- CLIN 1004-B (Enterprise 25-Node Installation / Base Fleet License): $2,450,000.00 / Year.

4. Intellectual Property & Data Rights Statement

This price schedule contains no classified or restricted proprietary markings. All 637 core C2 software modules and WORM schemas developed exclusively at private expense are retained by Humphrey Virtual Farms LLC under DFARS 252.227-7018 / FAR 52.227-14, with standard Government Purpose Rights granted for funded Phase-1 deliverables.